Do not write outside the box

0 1 This question is about Group 1 elements.

0 1 . 1 What are the Group 1 elements known as?

[1 mark]

Tick (✓) **one** box.

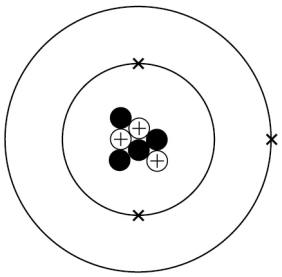
Alkali metals ☐

Halogens ☐

Noble gases ☐

0 1 . 2 **Figure 1** shows a lithium atom.

Figure 1



What is the number of electrons and neutrons in the atom of lithium?

[2 marks]

Number of electrons _____

Number of neutrons _____

0 1 . 3 What is the relative charge on a lithium ion?

[1 mark]

Tick (✓) **one** box.

+1 ☐ 0 ☐ –1 ☐

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0 1 . 4 Lithium is heated and then cooled in an experiment.

Lithium solid $\xrightarrow{\text{Stage } \mathbf{1}}$ Lithium liquid $\xrightarrow{\text{Stage } \mathbf{2}}$ Lithium solid

Two physical changes happen in the experiment.

Draw **one** line from each stage to the physical change that happens in that stage.

[2 marks]

Stage	**Physical change**
	Boiling
	Condensing
Stage **1**	
	Dissolving
Stage **2**	
	Freezing
	Melting

Question 1 continues on the next page

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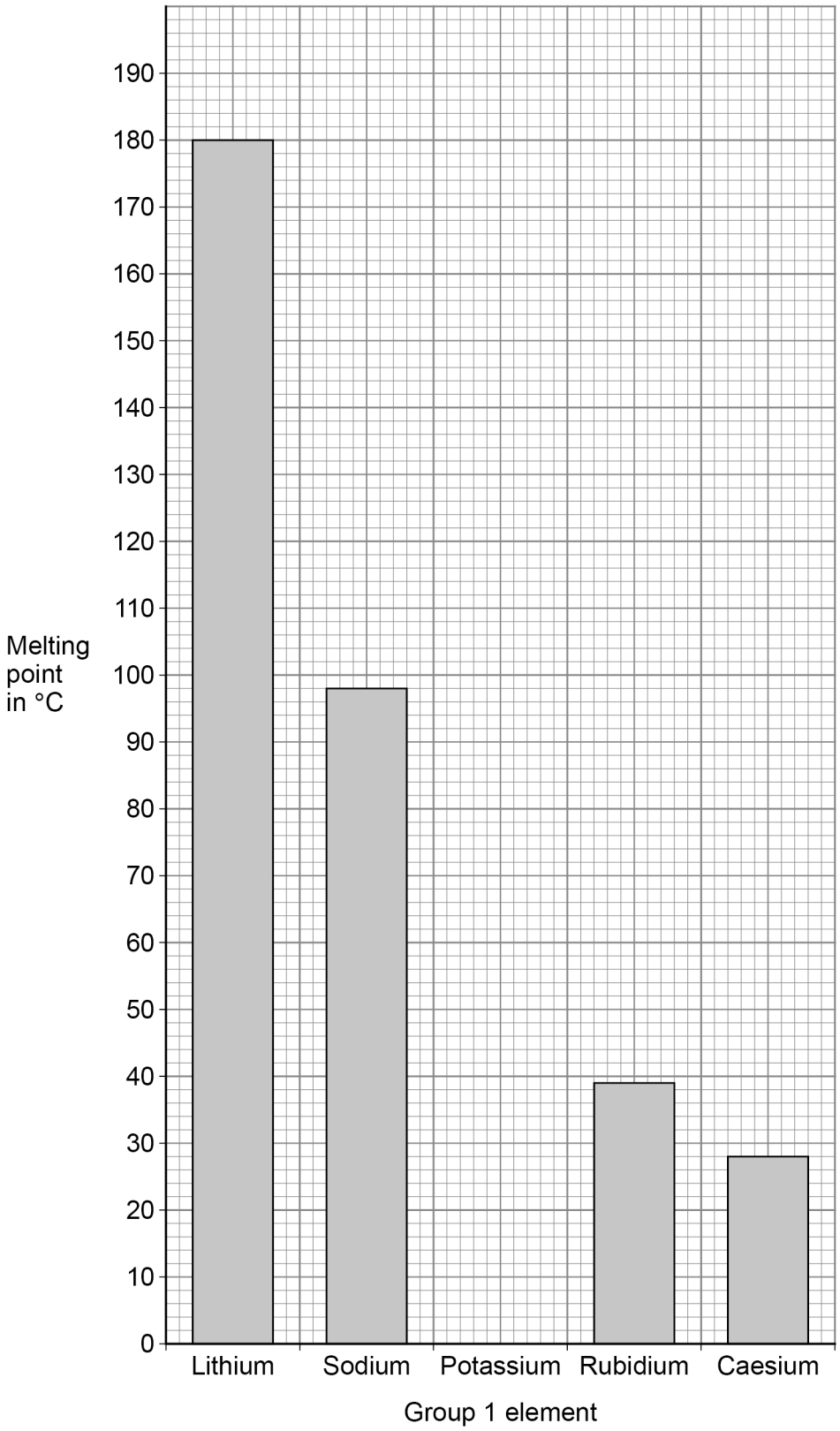
Figure 2 represents the melting points of some Group 1 elements.

Figure 2



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0 1 . 5 What is the melting point of caesium?

Use **Figure 2**.

[1 mark]

Melting point = _____ °C

0 1 . 6 The melting point of potassium is 63 °C

Draw a bar for the melting point of potassium on **Figure 2**.

[1 mark]

0 1 . 7 Describe the trend of the melting points of the Group 1 elements in **Figure 2**.

[3 marks]

_____

_____

_____

_____

_____

_____

0 1 . 8 The boiling point of sodium is 883 °C

What is the state of sodium at 150 °C?

Use **Figure 2**.

[1 mark]

Tick (✓) **one** box.

Gas ☐

Liquid ☐

Solid ☐

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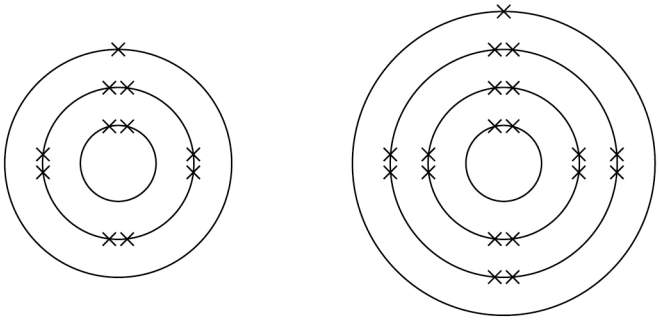
0 1 . 9 **Figure 3** represents the electronic structure of a sodium atom and of a potassium atom.

Figure 3



Sodium atom Potassium atom

Complete the sentence.

Choose the answer from the box.

[1 mark]

gains an electron	**loses an electron**	**shares an electron**

Potassium is more reactive than sodium because potassium more

easily _____ .

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